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Inventory in Times of War

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Abstract. Using data from 38,916 businesses in war-torn Colombia and from 5,138 attacks by the two rebel groups, FARC and ELN, we study how firms manage inventory during civil war. We obtain exogenous variation in the conflict intensity via a difference-in-differences model, which hinges on the peace process between Colombia's government and FARC. Relying on this identification strategy, we hypothesize and show that war causes two effects on firm-level inventories. First, it leads firms to replace physical assets (inventory) with fungible assets (cash), causing them to operate with an over-secured financial buffer, but a fragile operational buffer. Second, this inventory reduction occurs mostly in unprocessed inventories (finished-goods inventories are insensitive to violence), meaning that, although war-torn businesses are equipped to fulfill planned orders, they become inflexible at handling uncertain future demand. We then show that the magnitude of these effects is highly contingent on the firm's position in the supply chain, its proximity to distribution markets, and the type of attacks it is subject to. We then propose policies to address war-related risk in supply chains.

History: Accepted by Vishal Gaur, operations management.

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Keywords: inventory management • disaster and conflict management • causal inference models • empirical operations management

1. Introduction

Armed conflicts embroil 60% of the developing world. Forty-nine countries suffer from organized violence today, and, since 1989, there has been a yearly average of 20 concurrent civil wars waged. In these countries, warfare sabotages commodity production: hydrocarbons in South Sudan, steel in Libya, sugar in Somalia, and lumber in Myanmar. And bloodshed in a developing country disrupts supply chains, the very economic lifeblood of developed nations. What remains unknown is how war disrupts plant-level operations—how farmers plan their harvests when conflict looms near plantations, how firms manage raw materials in a combat region, or how warfare affects production in the factory.

This paper studies how firms manage inventories during armed conflict. But why should war affect inventory? Simply put, because managing physical stock in a war-ravaged region is a difficult task. Storing inventory becomes riskier because rebel groups, whose livelihood comes from looting, target inventory-rich firms. And inventories may be destroyed or rendered unusable when infrastructure is vandalized—for instance, when rebels bomb the town's power plant and leave its factories without electricity. Inventory

transportation also becomes riskier because rebels often block roads to steal cargo.

We combine data from the ongoing Colombian civil war, one of the longest-lasting of such conflicts, to compile a 16-year panel data set of 38,916 firms in war zones (including plantations, livestock farms, mills, factories, and the like). From such data, we note each establishment's geographic coordinates, accounting metrics, and inventory (raw, work-in-process, and finished goods). For instance, we track (i) the amount of sugarcane planted by a sugar refinery, (ii) the amount of sugar being refined in the factory, and (iii) the packaged/ready-to-sell sugar stored in the firm's warehouse. We also examine data on 5,138 attacks, observing each event's date and whereabouts, the perpetrator(s), the type of aggression, the target, and the number of injuries and deaths.

By merging these data, we measure the conflict intensity experienced by each firm and how each adjusts its inventories in response. But this raises another issue: The conflict intensity and a firm's inventory are simultaneously affected by unobserved factors. For instance, an increase in a crop's price will cause firms to, accordingly, increase the acreage of crops planted and the amount stored in their factories.

But this hike will also lure rebels to target regions that produce the higher-priced crops.

We find exogenous variation in the conflict intensity via a “natural experiment” from the Colombian civil war, which had been traditionally waged by two antigovernment groups: the Armed Revolutionary Forces of Colombia (FARC) and the National Liberation Army (ELN). For decades, the government attempted, but failed, to negotiate with both groups. But, finally, in late 2012, FARC agreed to initiate a peace deal. What made this agreement possible was that, months before, two of FARC’s leaders had died, triggering a leadership change. FARC’s new leadership shifted from a military to a political orientation, sought to stop warfare, and aimed to become a political party. ELN’s leadership, on the other hand, remained militarily oriented. After months of confidential negotiations, FARC announced a roadmap to end the conflict and de-escalated its violent activity, which led to a sudden reduction in warfare in its territories, whereas ELN’s territories continued to be embroiled in war. This event allowed us to use a difference-in-differences (diff-in-diffs) model with the treatment group comprising firms in FARC-dominated territories and the control group comprising those under ELN siege.

Relying on this identification strategy, we hypothesize (and show) how war affects inventories. Our theoretical framework departs from that of Riddick and Whited (2009) and Gao (2018), who posit that firms hedge risk by either accumulating fungible assets (i.e., cash) or physical assets (i.e., inventory). Fungible and physical assets—both labeled as working capital—serve as buffers against financial and operational frictions, respectively. The optimal working-capital allocation balances out each component’s marginal value; firms will funnel extra investments into inventory if the operational buffer is more fragile and into cash if the financial buffer is more fragile.

Applying this framework to the data, we demonstrate two effects, namely, (i) the *working-capital substitution effect* and (ii) the *lean-inventory effect*:

1. *The working-capital substitution effect.* War shifts the optimal working-capital allocation, causing firms to funnel resources away from inventory and invest more in fungible assets. Specifically, in reaction to war, Colombian firms decreased their inventory-to-assets ratio by 9%–12% and increased their cash-to-assets ratio by 7%–11%. The reason is that war, by increasing the cost of holding physical stock, makes it a costlier form of working capital. A reduction in the value of inventory, in turn, increases the relative value of fungible assets, which are protected in the banks’ vaults—and, thus, shielded from war. Accordingly, war makes firms operate with an “oversecured” financial buffer (by accumulating excessive fungible

assets) at the expense of having a fragile operational buffer (by having inefficiently low inventories).

2. *The lean-inventory effect.* The inventory retrenchment occurs mostly in raw and work-in-process inventories; finished-goods inventories are relatively insensitive to violence. For example, although war would induce a grain plantation to reduce its *unprocessed* grain by 12%, it would only cause the plantation to lower its *ready-to-sell* grain by 3%. During interviews, managers from war zones explained that, in the absence of war, firms produce excess unprocessed inventory to improve their reaction time to unexpected demand. For instance, a biodiesel producer would normally store extra raw corn because, if demand unexpectedly rises, the producer could quickly turn the corn into biodiesel. But violence, making such stockpiling much riskier, discourages managers from holding inventories for this purpose. In this fashion, war causes firms to produce enough to precisely fulfill planned orders and not to risk preparing in advance for possible future orders.

In spite of these results, it would be a stretch to claim that war affects all firms the same way or that firms react identically to all forms of violence. Surely, we cannot make the same inferences about the inventory of a manufacturer and that of a retailer, and it would be disingenuous to assume that a road blockage and a town takeover have the same effect. Thus, we explore two extensions: (i) firm-level moderator effects and (ii) attack-level moderator effects.

To develop our first extension (firm-level moderators), we identify two factors that characterize how firms manage inventories and then how these moderate our results. The first factor is the firm’s position in the supply chain (which dictates the nature of the inventories); the second factor is the firm’s proximity to distribution markets (which dictates its ability to acquire inputs and deliver output). We arrive at the following results:

1. *The supply chain effect.* In terms of timing, downstream firms adjust their inventories immediately in reaction to war, whereas upstream firms do so more slowly. But in terms of magnitude, upstream firms reduce inventories more abruptly than do downstream firms. The timing disparity is explained by Fackler and Livingston (2002), who note that commodity producers—those in the supply chain’s upper tier—have longer production lead times because they “grow” inventory from scratch (unlike downstream firms, who source inventories from upstream). The disparity in the adjustment’s magnitude is explained by two factors: (i) the unusually large-scale economies that characterize commodity production, which means that inventory adjustments are made in large batches; and (ii) the inherent vulnerability of commodities to war-related crime because these commodities,

unlike downstream inventories, are stored in the wild and are thus more easily accessible to marauders (Fearon 2005).

2. *The market-access effect.* Firms located near a distribution hub reduce inventory more aggressively; we find that an extra 100 km of distance, on average, moderates this reduction by 4.8%–5.3%. This finding is explained by the Economic Order Quantity (EOQ) model, which benchmarks inventory shipping costs and carrying costs. That is, when a firm is near a distribution hub, inventory travels less and, as a result, is less vulnerable to road attacks. In this case, war increases only the firm's holding cost, the EOQ denominator, but not its shipping cost, the EOQ numerator, making the inventory reduction sharper. But when a firm is far from a hub, inventory is more vulnerable to road attacks. In this case, war affects both the firm's shipping and carrying cost, the EOQ numerator and denominator, lessening the inventory reduction.

In our second extension (attack-level moderators), we show that firms react in opposite ways to private and public infrastructure attacks. In particular, they reduce inventory when the rebels terrorize nearby commercial establishments, but stockpile after a nearby attack on public infrastructure. We also find that businesses are more sensitive to the economic harm inflicted by an attack than they are to the human harm caused by violent attacks.

2. Literature Review

We first review two questions from the political science literature: What is war, and what causes it? Next, we look at the macroeconomics and microeconomics literature, which, respectively, examine how war affects countries and businesses. We then zoom in to the operations-management literature, which investigates how war affects businesses' internal operations.

2.1. War and Political Science: What Is War—and What Causes It?

The “violent taking of human life is the primary and dominant characteristic of war,” Small and Singer (1982, p. 206) argue, adding that wars must also involve “organized armed forces... and a minimum of 1,000 battle-related fatalities within a 12-month period.” But this definition lumps together more than 400 conflicts into the same category, which has led political scientists to differentiate wars in terms of location, intensity, and conduct.

In terms of *location*, there are interstate wars (those waged between two recognized states) and civil wars (those fought between the military and civil militia within a territory). As for *intensity*, there are total wars (which are unrestricted in terms of resources) and limited wars (those in which the contenders do not

expend all resources on fighting). For *conduct*, wars can be conventional or unconventional, depending on whether the sides respect international warfare treaties—but if the enemies use different tactics or have disparate strength (e.g., guerrilla warfare), wars can be asymmetric.¹

A natural question to ask is, what causes war? And, almost invariably, researchers find that wealth, or lack thereof, is by far the most robust driver. Assembling macro-level data on 27 wars, Collier and Hoeffler (2002) find a “substantially large and negative correlation” between gross domestic product (GDP) and conflict, while Justino (2007) finds that war is linked to low per capita income and sluggish economic growth. Holding wealth constant, countries are prone to war when they have weak institutions, mountainous terrain (Blattman and Miguel 2010), commodity-reliant economies (Dube and Vargas 2013), and rebels with easy access to financing (Collier and Hoeffler 2004).

2.2. War and Macroeconomics: How Do Conflicts Affect Nations?

Whereas nations don't immediately internalize the effects of war—Humphreys (2003), in fact, finds that a country's income is stable during conflict—neighboring regions are affected by negative spillovers, such as domestic destabilization and refugee flows (Murdoch and Sandler 2002, p. 92). But, as Murdoch and Sandler (2002) note, everyone suffers in the long run, adding that “civil wars have significant, but modest, negative influences on the steady-state level of GDP per capita [both in the host and neighbouring countries].” War also hampers a country's long-term domestic investment (Imai et al. 2000), depreciates physical capital, impairs human capital (Humphreys 2003), and disrupts national production networks (McDougal 2017).

2.3. War and Microeconomics: How Do Conflicts Affect Firms?

With the intent of understanding how war affects businesses, microeconomists dissect aggregate economic measures into various microcomponents. Klapper et al. (2013), for example, examined a business census in Ivory Coast to show that war cripples firm-level investments. Amodio and Di Maio (2017) used survey data from the Palestinian Central Bureau of Statistics to discover that civil war distorts input access. Using survey data from the 2008 Russia–Georgia conflict, Petracco and Schweiger (2012) show that conflicts decrease payroll size. And Collier (1999, p. 180) splits economic activities into war-invulnerable and war-vulnerable sectors and then shows, using 259 industry-level data points from Uganda's civil war, that there is a recomposition in GDP across these sectors, mentioning that, during conflict, “the war-invulnerable activity nearly doubled as a share of

GDP whereas...the share of the war-vulnerable group of activities nearly halved.”²

2.4. War and Operations Management: How Do Conflicts Affect the Internal Operations of a Firm?

War is not a foreign concept to operations management researchers. The field itself, in fact, is a product of war:

All serious accounts of the origins of Operations Research agree that the terminology was initially applied in Britain just prior to World War II ... [when] the military services began assembling scientists, engineers and mathematicians ... to conduct research directly in support of high-level officers (Thomas 2015).

With World War II long past, the field now examines other aspects of war, such as terrorism prevention. Zhuang and Bier (2007, p. 989), for example, find that “when increased defensive investment causes the attacker to redouble his efforts, defensive investment against terrorism will not decrease the probability of a successful attack.” Pinker (2007) finds that broadcasting public terrorism warnings can deter attacks, but that private warnings (issued to security forces) are more effective when the attack’s timing is certain. Bakshi and Gans (2010) show, via a game-theory model, that the Customs-Trade Partnership Against Terrorism program can be made more efficient with increased private-sector security processing.

Operations research also studies humanitarian operations during wartime. Jola-Sanchez et al. (2016), for example, analyze 163 Colombian hospitals and conclude that war makes health centers 11% more productive, but 10% less cost-efficient. Eftekhari and Van Wassenhove (2016) study how conflicts affect last-mile fleet management in humanitarian operations, and Pedraza-Martinez and Van Wassenhove (2013) show that conflict intensity does not affect the salvage value of humanitarian vehicle fleets.

3. Theoretical Framework: War and Inventory Management

Despite the large inflow of research on civil war, few have studied how war affects plant-level operations. This limitation might be because the most adept war researchers—political scientists, public policy researchers, and economists—entertain other questions or, simply, because there are not much data available. No matter the reason, today we have (at best) gleaned only rough insights about the day-to-day operations of war-ravaged businesses.

We contribute to the literature by employing a microscopic view to bring into focus the effect of

conflict on inventory management. We first discuss, in Section 3.1, the mechanisms through which civil war affects inventory. Then, in Section 3.2, we formulate two hypotheses on how war alters firm-level inventories.

3.1. Why Does War Affect Inventory?

Inventory-management textbooks define three cost components pegged to inventory: (i) acquisition cost, (ii) carrying cost, and (iii) shortage cost.³

3.1.1. The Acquisition Cost. This component includes costs associated with bringing inventory to the factory. And anecdotal evidence shows that war makes this task difficult. Rebels hijack transportation infrastructure, block roads to steal cargo, charge illegal tolls, and kidnap drivers for ransom money. The Global Terrorism Database, which documents 180,000 war-related attacks worldwide, reports upwards of 8,000 attacks targeted at bridges, highways, railways, airports, and seaports. The database also reports tens of thousands of attacks directed at cargo planes, trucks, and ships. We offer four examples taken directly from this database:

“FARC dynamite attack destroys bridge in Huila Department, Colombia.” (May 7, 2002)

“[Boko Haram] assailants blew up a bridge in Gada Mboulo village, extreme-north region, Cameroon.” (July 23, 2014)

“National Liberation Army (ELN) rebels set up fake roadblocks on the bridge over Calderas River in San Luis Municipality, Colombia, on the road between Bogota and Medellin many buses were burned and the perpetrators fled with a truck full of cattle.” (September 18, 1999)

“An explosive device detonated on railway tracks in Bakhtiarabad area [Pakistan] ... the railway tracks were damaged in the blast. The Baloch Republican Guards (BRG) claimed responsibility for the incident.” (February 7, 2015)

Firms pay hefty prices to avoid these attacks by hiring security to transport inventory, choosing inefficient routes in search of safety, or procuring costlier transportation modes (such as airfreight).

3.1.2. The Carrying Cost. Carrying inventory is risky because rebels subsist by looting supplies from firms, which is often their main funding source (Humphreys 2003). Businesses also lose inventories when they suffer direct attacks, such as bombings, or because inventory spoils after a public infrastructure attack, which causes electricity, water, or communication outages in the region. The Global Terrorism Database reports 64,181 attacks directed toward businesses and

9,043 attacks on public services infrastructure. Below are four examples:

“New People’s Army guerrillas raided the compound of the SR Metals Mining Corp., the largest nickel ore mine firm in Tubay Town, Philippines The attackers poured gasoline on three backhoes and set these on fire the mining firm’s refusal to give revolutionary tax to the NPA might have motivated the attack.” (December 21, 2008)

“A Coca-Cola factory was bombed in Cali, Colombia, by ELN assailants.” (April 19, 1998)

“In Dakwa, Orientale, Congo armed assailants attacked the village . . . killing livestock and looting and burning a health center, several residences, pharmacies, retail shops, and a Catholic mission.” (June 2, 2009)

“As part of a series of nationwide infrastructure attacks, Fuerzas Armadas Revolucionarias de Colombia (FARC) members blew up the power substation in the city of Arauca, Colombia, which resulted in wide-scale power outages and water service disruptions throughout the country.” (February 27, 2002)

3.1.3. The Shortage Cost. This component includes costs associated with emergency sourcing fees or the opportunity cost of lost sales. And civil war can affect this component in two ways. On the one hand, it can increase the likelihood of a shortage, namely, by increasing demand variability or by making emergency sourcing unavailable (e.g., if rebels block access to roads). On the other hand, it can affect the shortage cost itself, that is, by increasing emergency sourcing fees (e.g., via increased transportation rates).⁴

3.2. Hypotheses: How Does War Affect Inventory?

The arguments presented above show that war is unlike other operational disruptions in that the likelihood (and the intensity) of plunder increases with the amount of inventory held—provisions attract marauders like honey attracts bears. And this is what differentiates war from other risks. A hurricane isn’t more likely to strike when oil drillers choose to stockpile more barrels; a labor strike doesn’t intensify just because miners choose to transport more copper. But attacks are more likely, and they hit harder, when the barrels are overflowing with crude oil and when copper-laden trucks travel more frequently. Thus, although firms can hedge against the risk of natural disasters, labor disputes, or supplier bankruptcies by holding inventories, they cannot hedge against the risk of war by doing so, because the very act of holding inventories affects the likelihood and magnitude of an attack.

How can firms deal with war-related threats? To answer this question, we formulate a theoretical framework inspired by both Riddick and Whited (2009) and Gao (2018). In this framework, firms use working capital to deal with day-to-day uncertainty risk. There

are two types of working capital: fungible assets (i.e., cash) and physical assets (i.e., inventory). Fungible assets are used as a buffer against financial risk, whereas inventory is used as a buffer against risks such as demand uncertainty or supply disruption.

Firms determine their optimal working-capital allocation by deciding how much to feed into their financial buffer and into their operational buffer. To determine this allocation, Gao (2018, §3.5) formulates an infinite-horizon inventory supply model and finds that the optimal working-capital ratio is a function of the net-present value offered by each component. Thus, firms will funnel additional investments into inventory when the operational buffer is more fragile, and vice versa.

Because war increases the cost of acquiring and carrying physical stock, it makes it costlier to use inventory as a risk-management buffer. An increase in the cost of physical inventory will, accordingly, increase the relative value of fungible assets—which are protected from the rebels in the vaults of financial institutions—thereby making it a relatively more desirable form of working capital. Thus, we conjecture that an increase in conflict prompts firms to funnel investments away from inventory and into more fungible assets. Put another way, war creates an environment whereby firms operate with an over-secured financial buffer (by hoarding excessive fungible assets) at the expense of having a fragile operational buffer (by having an inefficiently low amount of inventory). We label this effect the working-capital substitution effect.

Hypothesis 1 (Working-Capital Substitution Effect). *When conflict intensity is high, firms in war-torn areas will divert resources away from inventory and into fungible assets.*

In the process of reducing inventories, firms decrease exposure to war-related risk, but increase exposure to day-to-day operational uncertainty. How do they unravel this dilemma? Sheffi (2001) proposes disentangling the risks associated with normal business uncertainty and those associated with war. This means that firms should have a dual inventory system to tackle each uncertainty source by keeping a lean inventory system with enough finished inventories, but without excess safety stock. This strategy minimizes exposure to day-to-day uncertainty and expected shortage costs by keeping a strategic inventory system to tackle civil-war-specific disruptions. Although essential inputs and finished inventory are included in this system, raw inventory, work-in-process, and speculative inventory should be minimized, thereby curtailing the stock that ends up embroiled in the crossfire.

Anecdotal evidence shows that firms in war zones behave as suggested by Sheffi. After interviewing businesspeople from conflict-ridden zones, a Colombian

logistics manager (who sources from war-ravaged suppliers) impressed upon us the following insight:

Most firms [in conflict zones] are small and keep low inventory. These businesses keep a minimum inventory level because they are exposed to attacks; security is a big deal, and businessmen cannot risk their capital. My experience has been that our local suppliers . . . keep minimum low-rotation inventory. However, they keep adequate levels of finished, high-rotation, inventory. We purchase from these firms regularly—so they don't want to let us down.

We also learned that commodity producers (e.g., crop plantations) have significant economies of scale and low variable production costs. In normal conditions, these firms would produce in excess of demand—for example, a farmer would store unfinished grain to hedge against unexpected demand fluctuation. But warfare makes this practice riskier: Factory attacks and infrastructure bombings jeopardize inventories and increase inventory-holding costs. Accordingly, civil war should induce firms to produce just enough to precisely fulfill planned orders and not risk preparing in advance for possible future orders. We refer to this as the lean-inventory effect.

Hypothesis 2 (Lean-Inventory Effect). *Most of the inventory reduction will materialize in the form of early stage inventories.*

To test these two hypotheses, we develop an empirical framework that uses data from the Colombian civil war, which we explain next.

4. Empirical Framework

4.1. Background: The Colombian Civil War

In 1964, a civil war began between two Marxist-oriented groups, FARC and ELN, and the Colombian government. Other factions—EPL, M-19, and right-wing paramilitary groups—joined the conflict, but, since then, they have either disbanded (decades ago) or, today, hold innocuous military strength.

In the last two decades, the conflict caused 1 million deaths, 8 million displacements, and thousands of attacks from FARC and ELN. Both groups use similar military tactics to strike the population: bombing infrastructure, wrecking factories, blocking roads, destroying bridges, and burning vehicles and cargo (see Figure 1). These attacks also allow the groups to fund their operations. Striking businesses allows them to loot inventory, hijacking roads allows them to snatch merchandise in transit, assaulting civilians enables them to demand ransoms, and peacocking their military might empowers them to levy “revolutionary taxes” on firms. Moreover, such pilfered inventories are used to recruit and feed combatants.

Although the war has extended almost everywhere throughout Colombia's rural regions and small towns, it has barely crept into big cities like Bogota and Medellin (apart from one-off attacks) or inhospitable places like the inner Amazon rainforest. Figure 2 shows that FARC dominates Colombia's south and west, whereas ELN dominates the northeast and central regions. The two groups seldom operate in the same zone, except for strategic enclaves, such as the frontier bordering Venezuela. In fact, our data show that only 7% of the territory is shared. This tactic gives the groups territorial exclusivity, allowing them to channel their forces against the government instead of using combatants to attack each other.

4.1.1. The Peace Process. For decades, the Colombian government has sought peace agreements with rebel groups. And, whereas in the 20th century, it succeeded in brokering agreements with smaller groups, FARC and ELN repeatedly refused to disarm due to their diehard military orientation. But FARC changed its mindset after its two commanders died and were replaced by politically oriented leaders. Under new leadership, FARC began secret exploratory talks with the government in Norway and Cuba, in secluded areas to prevent press leakages and political interference. And on November 19, 2012, the government publicly announced the beginning of a peace process with FARC—an unexpected announcement given 50 years of conflict.

During the process, the government and FARC kept military guard over the territory and did not permanently stop military operations. Both parties, however, ceased fire as a sign of good faith. And our data show irrefutable evidence of this goodwill: Attacks and casualties by FARC had decreased 98% by 2016. Meanwhile, ELN continued its usual military activity and became the main source of attacks in Colombia. However, ELN continued to respect FARC's heavily guarded territories.

4.2. Data

To build our data set, we harmonized three types of data: (i) firm-level metrics, (ii) war-level metrics, and (iii) municipality-level metrics.

4.2.1. Firm-Level Metrics. We obtained firm-level data from the Colombia Ministry of Trade's Business Census. Spanning the years 2001 to 2016, the census reports data on 38,916 firms and 259,448 firm-year observations. We observe each firm's legal identifier, location, and over 150 accounting and operational metrics (equity, cash, assets, revenue, and the like). We also observe all stages of the firm's inventory: raw material, work-in-process, finished goods,

Figure 1. Exhibits of the Colombian Civil War



Notes. (First row) Exhibits showing FARC and ELN army members. (Second row) Exhibits showing how FARC and ELN mark territory after a town takeover. (Third row) Exhibits showing a factory and a business district after an attack. (Bottom row) Exhibits showing a fleet of cargo-laden trucks in the aftermath of FARC and ELN highway ambushes.

and total inventory. Table 1 shows sample rows of a coffee-processing business. All statistics are in constant-value Colombian currency, to account for inflation.

A potential concern is that firms underreport assets to avoid being targeted by rebels. The ministry, however, keeps the census confidential on its servers, which we could only access via a scientific data-access request. In addition, the ministry audits firms, and misreporting leads to fines, jail time, or the establishment's closure.

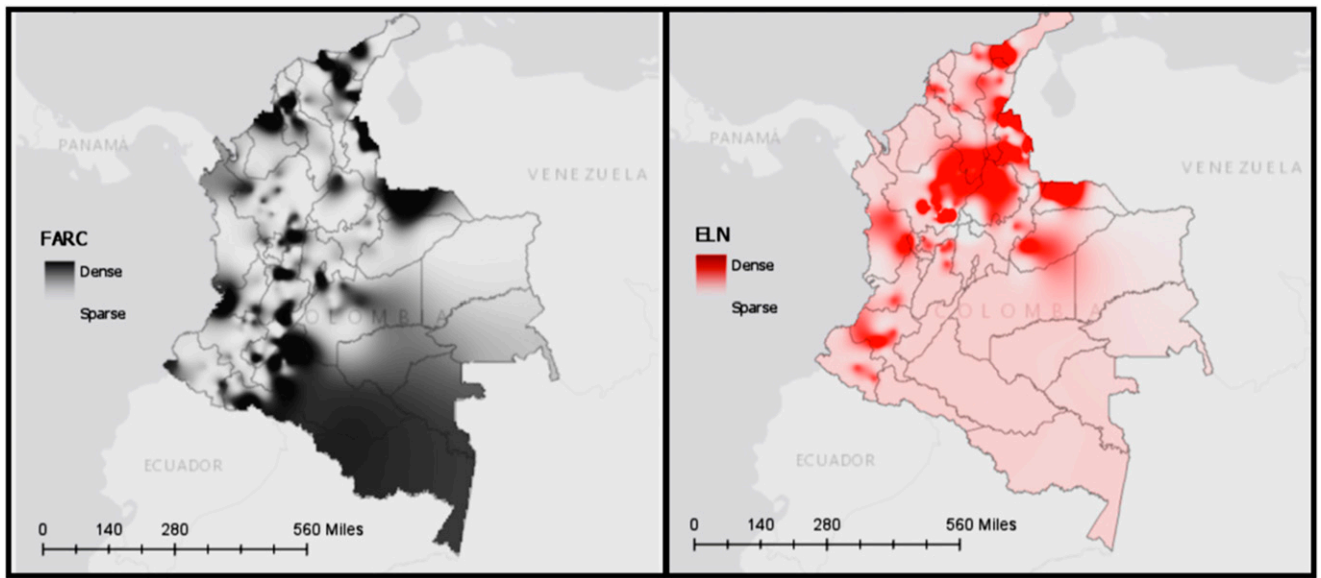
4.2.2. War-Level Metrics. We assemble attack-level data from Colombia's Ministry of Social Prosperity's "Memoria Historica" data set. Memoria Historica is a government census covering every reported attack made by FARC or ELN between 1998 and 2016. Each row includes information about an attack's type, date, coordinates, injuries, deaths, at-fault rebel group,

target, property damage caused, and text descriptions about the incident. Table 2 provides entries on five sample attacks.

A potential concern is that some assaults are not reported because civilians are afraid to denounce rebels. Colombia's army, however, runs media advertisements that promote anonymous hotlines and a website to report these incidents. In addition, the nature of war-related attacks—bombings, town takeovers, en-masse road blocks, massacres, and kidnappings—is such that they are unlikely to be glossed over. Insurgents also publicize attacks, to show off their military power, and, thus, invariably leave their signature on an attack's site by placing flags, graffiti, or stamps with their logo (see Figure 1).

4.2.3. Municipality-Level Metrics. We gather data on all 1,123 Colombian municipalities, merging (i) Colombia's

Figure 2. (Color online) Geographic Distribution of Attacks



Note. The shading density represents the conflict intensity.

Administrative and Political Registry and (ii) the University of Los Andes’ Geographic Data Set. The first describes the municipality’s natural geography, and the second reports 2,150 socioeconomic characteristics, between 1993 and 2016. From the second database, we also identify the country’s trade hubs and regional markets. Table 3 provides sample entries for four municipalities.

4.3. Sample Selection

Table 4 summarizes the datasets, which we merged to obtain a panel with 257,865 firm-year observations. Spanning the years 2001 to 2016, our panel records each firm’s coordinates, operational and accounting metrics, municipality data, and data on all nearby attacks. For example, we observe that Agricola el Chaco S.A., a rice producer, has coordinates 4.28, –75.58. Within a 50-km radius, FARC made 10 attacks between 2001 and 2009, including a vehicle-torching assault, with no casualties; a military building bombing,

with eight casualties; and an ambush, with one casualty. ELN waged no attacks in this region.

In our main analysis, we use three years of data in the preannouncement phase (i.e., 2010 to late 2012) and three years in the postannouncement phase (i.e., 2013 to late 2016). We select this time span because in 2010, the Colombian government held general elections. Thus, we restrict our sample to include observations under the same government to reflect a consistent political environment (an assumption that is critical in diff-in-diffs models). We also log-linearize all continuous variables to account for their positive and skewed nature and remove outliers by winsorizing our sample.

5. Identification and Results

5.1. Identification Strategy

It is difficult to obtain causal estimates of the relationship between inventory and the conflict, because there are unobserved shocks that simultaneously

Table 1. Firm-Level Data: Sample Entries

Municipality	Coordinates	Name	Year	Assets		Equity	Cash	Revenue	Net Income	Inventory			
				Total	Current					Raw	WIP	Finished	...
Caicedonia	–75.83, 4.33	Sociedad	2010	13,063	604	12,588	106	671	–192	3	239	116	...
		Promotora	2011	15,158	1,416	14,614	68	104	130	5	264	103	...
		Agro-	2012	15,230	1,508	14,764	61	151	261	7	324	85	...
		industrial	2013	15,566	2,145	14,632	247	8	864	9	330	41	...
		Cafetera	2014	15,954	2,179	15,129	102	91	423	11	253	73	...

Notes. The variables *Assets*, *Equity*, *Cash*, *Revenue*, *Net Income*, and *Inventory* are logged in thousands of Colombian Pesos (COP). In June 2019, US\$1≈COP\$3400. Geographic coordinates are in decimal degrees and in terms of longitude and latitude. WIP, work-in-process.

Table 2. Attack-Level Data: Sample Entries

Perpetrator	Coordinates	Date	Attack type	Details	Target	Weapons	Killed	Injured
FARC	−70.75, 7.08	02/27/02	Public infrastructure	Bombing	Utilities	Explosives	0	0
FARC	−75.53, 10.41	07/18/02	Private property	Bombing	Citizens	Explosives	3	16
FARC	−74.20, 11.24	11/19/02	Commercial property	Bombing	Business	Explosives	1	5
ELN	−75.33, 6.43	05/02/02	Roads and cargo	Assault	Bridge	Incendiary	0	0
ELN	−75.56, 6.24	07/23/02	Commercial property	Bombing	Business	Vehicle	2	13

Notes. Geographic coordinates are in decimal degrees and in terms of longitude and latitude. Date format: MM/DD/YY.

affect these two factors. If a crop's price were to increase, for example, managers would increase supply and, accordingly, adjust inventories. Meanwhile, a rise in the crop's price would make rebels more likely to target the crop's producers.

To obtain exogenous variation in the conflict level, we use the announcement of the peace process between Colombia's Government and FARC, in late 2012, as a quasi-exogenous event. This event is ideal for three reasons. First, it had a consequential effect on the level of conflict. Figure 3 shows that FARC's attacks plummeted in the wake of the process (whereas ELN's attacks continued to follow the same trend). Second, the event was sudden and unanticipated because, before the announcement, the two parties held "secret exploratory talks in Havana ... [with] confidentiality as one of their key principles" (Norwegian Centre for Conflict Resolution 2018). The third reason is that the shock affected only a fraction of Colombia's territory because, as our data reveal, FARC and ELN had territorial monopolies. Thus, we can clearly define the treatment and control groups—firms operating in FARC- and ELN-dominated regions—and compare inventory changes across both groups.

5.2. Identifying the Treatment and Control Groups

We follow two steps to split firms into the treatment and control groups.

Step 1. *Splitting firms into war-torn and peaceful regions.* We count the battle-related deaths within 50 km of each firm. We stipulate that a firm operates in a peaceful region if, in the 10 preceding years, there were fewer than d battle-related deaths per 100,000

inhabitants. But what constitutes an appropriate value of d ? The conflict-intensity distribution (in Online Appendix IV, Figure A.9) indicates a bimodal distribution, which means that there is a patent division between regions with no war (63.67% witnessed no attacks) and those afflicted by it. For robustness, however, we obtain estimates using three war thresholds: $d = 5$ (low), $d = 25$ (medium), and $d = 50$ (high).

Step 2. *Splitting firms into FARC-dominated and ELN-dominated regions.* Focusing on the subsample of war-torn firms, we count the attacks made by FARC and ELN within 50 km of each firm's location. A firm belongs to the treatment group if FARC's attack count is larger and to the control group if ELN has the larger count. We classify 45,139 and 22,850 observations in the treatment and control groups, respectively. Tables 5 and 6 show summary statistics for municipalities and firms in both groups.

5.3. Models and Estimates

5.3.1. The Working-Capital Substitution Effect. To test our first hypothesis, we begin by measuring the effect of war on inventory, via the following diff-in-diffs specification:

$$INV_{it} = \alpha_t + \beta FARC_i + \rho AFTER_t + \tau(FARC_i \cdot AFTER_t) + \delta X_{it} + \varepsilon_{it}, \quad (1)$$

where INV_{it} is firm i 's inventory-to-assets ratio in period t , term α_t is a vector of time fixed effects, $FARC_i$ indicates whether the observation is in a FARC-dominated region, $AFTER_t$ indicates whether the

Table 3. Municipality-Level Data: Sample Entries

Code	Municipality	Coordinates	Altitude	Employment rate	GDP per capita	Illiteracy rate	Population	Poverty rate	Surface area	...
05002	Abejorral	−75.43, 5.80	2,125	0.97	7,307,526	0.24	20,566	0.47	497	...
50006	Acacias	−73.72, 4.01	522	0.94	8,506,859	0.18	54,134	0.36	1,149	...
68572	Puente Nacional	−73.68, 5.83	1,625	0.93	8,624,333	0.30	14,548	0.48	273	...
25779	Susa	−73.83, 5.44	2,586	0.98	6,930,469	0.30	9,861	0.39	111	...

Notes. Altitude is measured in meters above sea level. Surface area is measured in km^2 . Geographic coordinates are in decimal degrees and in terms of longitude and latitude.

Table 4. Data Set Summary

Provider	Database name	Time span	Observations	Unit of analysis	Variables	Sample variables
Colombian Ministry of Trade	Firm Reports SIREM	2001–2016	259,448	Firm-year	254	Name, Year, Municipality, Total Assets ...
Social Prosperity	Memoria Historica	1988–2016	5,138	Attack	13	State, Year, Municipality, Deaths, Attacks, Date ...
University of Los Andes	Municipality Panel	1997–2016	21,337	Municipality-year	2,150	Year, Population, Altitude, Literacy Rate ...
National Department of Statistics	Administrative Registry	N/A	1,123	Municipality	17	Municipality Code, Longitude, Latitude, Altitude, Surface ...

Note. NA, not applicable.

observation is recorded after the treatment date, and X_{it} is a vector of municipality- and firm-level controls.⁵

Finally, ε_{it} is an error term. We cluster all errors at the municipality level, following Angrist and Pischke (2008, p. 237), who argue that to avoid serial correlation one should “pass the clustering buck one level higher.” In our specification, the buck-one-level-up unit is the municipality.

Table 7 shows Equation (1)’s estimates across six specifications. In columns (I) and (II), we obtain estimates by using the “low” war threshold; in columns (III) and (IV), we use the “medium” threshold; and in (V) and (VI), we use the “high” threshold. Note that only the even-numbered columns include control variables and that we scale all coefficients by 10, for illustration purposes.

The diff-in-diffs coefficient is statistically significant across all specifications, showing that the conflict intensity and the inventory-to-assets ratio move in opposite directions. Furthermore, Figure 4 shows that this effect is driven by a change in inventory, the ratio’s numerator, as opposed to a change in assets, the denominator. In terms of magnitude, this change ranges between 9.6% and 12.1%.⁶

Having established the effect on the firms’ inventories, we now study the effect on their cash reserves. To this end, we regress the following equation:

$$CASH_{it} = \alpha_t + \beta FARC_i + \rho AFTER_t + \tau(FARC_i \cdot AFTER_t) + \delta X_{it} + \varepsilon_{it}, \quad (2)$$

where $CASH_{it}$ is the firm’s cash-to-assets ratio.

Table 8 presents Equation (2)’s estimates. And, as hypothesized, the firms’ cash and inventory levels move in opposite directions. Using the low-violence threshold (and our results from Tables 7 and 8), we find that an 11% reduction in inventory-to-assets ratio is met with a 7.6% increase in the cash-to-assets ratio; and, using the high-violence threshold, we find that a 12% reduction in the inventory-to-assets ratio is compensated for by a 11% increase in the cash-to-assets ratio. This means that a reduction in violence

causes not only a substitution in working capital, but also a reduction in the overall net working-capital investments.

5.3.2. The Lean-Inventory Effect. To test our second hypothesis, we re-estimate Equation (1), but instead of using total inventory as the dependent variable’s numerator, we replace it with (i) raw inventory, (ii) work-in-process (WIP) inventory, and (iii) finished inventory:⁷

$$RAW_{it} = \alpha_t + \beta FARC_i + \rho AFTER_t + \tau(FARC_i \cdot AFTER_t) + \delta X_{it} + \varepsilon_{it}, \quad (3)$$

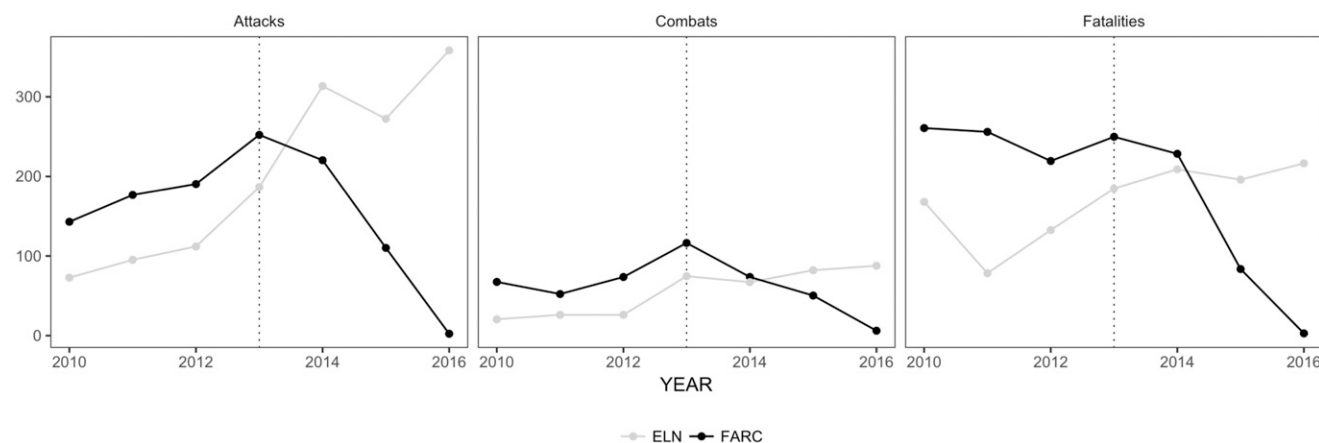
$$WIP_{it} = \alpha_t + \beta FARC_i + \rho AFTER_t + \tau(FARC_i \cdot AFTER_t) + \delta X_{it} + \varepsilon_{it}, \quad (4)$$

$$FINISHED_{it} = \alpha_t + \beta FARC_i + \rho AFTER_t + \tau(FARC_i \cdot AFTER_t) + \delta X_{it} + \varepsilon_{it}. \quad (5)$$

Table 9 shows that war causes managers to reduce all types of inventory—raw, work-in-process, and finished-goods. A palm-oil producer will store fewer raw fruit bunches, will not hold as much ready-to-process (deshelled) nuts, and will hoard less packaged-to-sell oil. But the inventory retrenchment is mostly reflected in unfinished inventories, where the coefficients are 2.5 times larger for raw and work-in-process inventories than for finished inventories. These differences in magnitude are statistically significant across all six specifications, thereby supporting our second hypothesis.

To summarize, our estimates provide compelling evidence of two effects: the working-capital substitution effect and the lean-inventory effect. With that said, we need to acknowledge the vast heterogeneity that characterizes our sample. To assume that war affects all firms in exactly the same ways is oblivious to the fact that our sample spans thousands of commodities—from livestock to natural gas to cotton—which are stored in different regions and conditions. Also, we need to acknowledge that the nature of war can vary tremendously. In some cases, firms are

Figure 3. War Trends



exposed to road attacks and commercial-property attacks; in others, they are more likely to experience murder and bloodshed.

To obtain a more nuanced picture about how these differences influence our results, we next explore two extensions: one that studies firm-level moderators and one that studies attack-level moderators.

6. Extension I: Firm-Level Moderators

From the literature, we identify two factors that affect how firms manage inventories and, subsequently, how these factors moderate the relationship between inventory and conflict. The first factor is the firm's position in the supply chain, which determines the nature of the inventories (Fackler and Livingston

2002, Fearon 2005). The second factor is the firm's proximity to distribution hubs, which dictates its ability to transport inventory to and from the factory (Amodio and Di Maio 2017).

6.1. Moderator 1: The Supply Chain Effect

Each firm holds a unique assortment of inventories, and when it comes to adjusting these inventories, they each encounter different hurdles. But to understand how these differences shape the relationship between inventory and conflict, we first need to classify inventories in a way that captures their idiosyncrasies. Supply chain-management researchers, who have taken on this task, have come up with a canny way to differentiate them—namely, by categorizing firm-level

Table 5. Summary Statistics by Firm (Before and After Treatment)

Variable	Group							
	FARC				ELN			
	Before		After		Before		After	
	Mean	St Dev	Mean	St Dev	Mean	St Dev	Mean	St Dev
Total Inventory	9.19	6.31	12.12	6.51	9.21	6.19	9.23	6.57
Finished Inventory	8.12	6.49	8.17	6.68	8.36	6.52	7.93	6.76
WIP Inventory	2.87	5.28	3.76	5.22	3.31	5.49	3.03	5.39
Raw Inventory	3.21	5.29	5.65	6.21	3.64	5.74	3.35	5.66
Cash	1.67	7.07	1.28	6.27	1.26	5.26	1.21	4.55
Equity	14.33	1.76	14.68	1.65	14.38	1.71	14.74	1.64
Assets	15.11	1.64	15.50	1.51	15.20	1.56	15.56	1.45
Net Income	11.44	2.33	11.74	2.35	11.55	2.31	11.90	2.23
Cost of Goods Sold	11.67	6.17	11.64	6.38	11.92	6.16	11.45	6.58
Profit Margin	−1.21	0.90	−1.16	0.88	−1.23	0.88	−1.16	0.89
Sales	14.77	2.01	14.96	1.99	15.00	1.90	15.08	1.89
Return on Assets	−3.62	1.36	−3.66	1.44	−3.62	1.39	−3.61	1.42
Financial Leverage	0.38	0.25	0.38	0.23	0.39	0.25	0.38	0.25
Revenue	12.69	3.80	12.91	3.82	12.82	3.93	12.84	4.11
Payroll	7.70	4.81	7.94	4.85	8.18	4.63	8.18	4.86
Firm-year observations	22,548		22,591		10,994		11,856	

Notes. All variables are logged in thousands of Colombian Pesos (COP). In June 2019, US\$1≈COP\$3400. GDP, gross domestic product; St Dev, standard deviation; WIP, work in progress.

Table 6. Summary Statistics by Municipality

Variable	Group			
	FARC		ELN	
	Mean	St Dev	Mean	St Dev
Altitude	1,030.95	893.85	1,217.86	771.85
Surface Area	813.34	1,539.11	838.40	401.45
Population	114.72	233.81	175.16	423.62
Poverty rate	0.53	0.15	0.47	0.15
GDP per capita	8.53	5.16	8.86	2.95
Gov. spending per capita	19.04	74.66	19.87	78.17
Homicides per 100k inhabitants	131.20	124.10	148.80	108.70
Illiteracy rate	0.17	0.08	0.14	0.07
Employment rate	0.89	0.06	0.86	0.10
Municipalities	144		264	

Notes. Altitude is measured in meters above sea level. Surface area is measured in km^2 . Population is measured in thousands of people. GDP per capita and Government (Gov.) spending per capita are measured in logged Colombian pesos (COP) (in June 2019, US\$1 $\approx$ COP\$3400). To create this table, we split peaceful and war-torn regions using a low war threshold. St Dev, standard deviation.

inventories into four sectors: commodity extraction, manufacturing, wholesaling, and retailing (Bray and Mendelson 2012). These four sectors define how firms acquire, store, and utilize their inventories: Commodity producers harvest inventories from scratch, manufacturers assemble these inventories into a final product, wholesalers distribute the final product, and retailers sell it.

We categorize our sample into these four sectors, using each firm's International Standard Industrial Classification (ISIC) code, from Colombia's Chamber of Commerce. When matching an ISIC code to a sector, we follow Bray and Mendelson's (2012, table 3) classification. After sorting out firms, we run two separate regressions. The first regression re-estimates Equation (1) across the four sectors.⁸ The second regression decomposes this effect across the time series,

via the specification of Autor et al. (2003), which adapts Granger's Causality test to a diff-in-diffs setting, such that:

$$\Delta_t INV_{ist} = \alpha_s + \underbrace{\Delta_t YEAR_{ts}}_{\text{time indicator}} + \underbrace{FARC_i}_{\text{treatment indicator}} + \underbrace{\omega \Delta_t X_{its}}_{\text{controls}} + \underbrace{\sum_t \beta_{its} \Delta_t YEAR_{ts} FARC_i}_{\text{year-by-year change in diff-in-diffs effect}} + \varepsilon_{it}, \quad (6)$$

where s subscripts the supply chain sector and Δ_t is a time-differences indicator ($\Delta_t x_t = x_t - x_{t-1}$). As Autor et al. demonstrate, this regression decouples the diff-in-diffs shock in the time series without introducing serial correlation. Thus, the β coefficients illustrate how the marginal effect lessens (or intensifies) across years.

The first regression estimates, shown in Table 10, demonstrate that the diff-in-diffs effect is statistically significant across all sectors. But the second regression estimates, shown in Table 11 and visualized in Figure 5, demonstrate that wholesalers and retailers immediately adjust inventories, whereas commodity producers exhibit a lagged reaction. In terms of magnitude, however, the adjustment is larger in the upstream portion of the supply chain.

Thus, when it comes to violence fluctuation, upstream firms react more slowly, but more forcefully, than do businesses further downstream. What explains these differences?

The difference in reaction timing is likely driven by one peculiarity that makes commodity producers unique: the way they acquire inventories. Unlike downstream firms, which source inventories from upstream, commodity producers grow inventories from scratch, harvesting spices, mining metals, and breeding animals. In fact, Fackler and Livingston (2002) show that commodity producers have extraordinarily long production lead times, which are

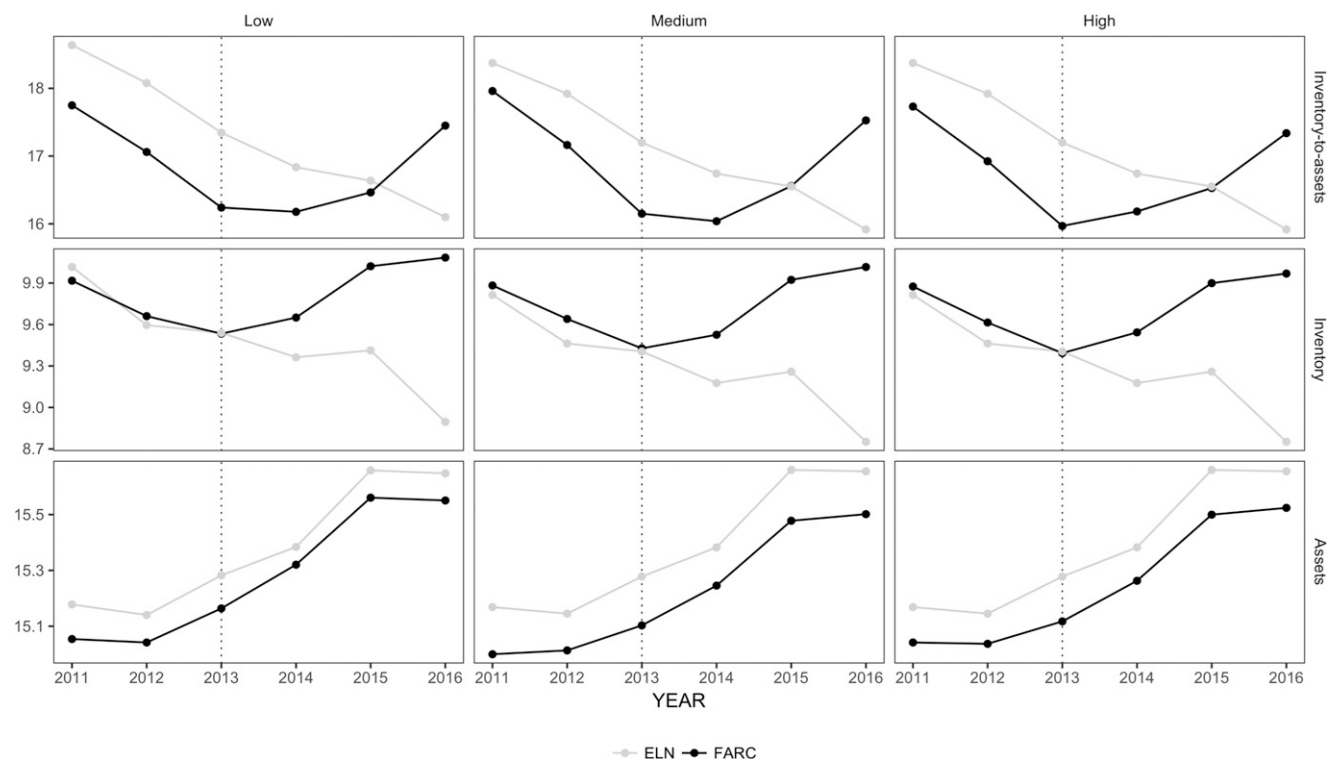
Table 7. Model Estimates: Inventory-to-Assets

Specification	I	II	III	IV	V	VI
FARC·AFTER	1.08*** (0.40)	1.10*** (0.40)	0.92** (0.46)	0.92** (0.45)	1.14** (0.46)	1.14** (0.46)
FARC	−0.97 (1.35)	0.92 (1.44)	−0.61 (1.41)	1.00 (1.52)	−0.89 (1.48)	0.76 (1.55)
AFTER	−1.41*** (0.21)	−1.41*** (0.21)	−1.34*** (0.26)	−1.34*** (0.26)	−1.34*** (0.26)	−1.33*** (0.26)
Dependent variable	$\log(\text{inventory-to-assets})$					
War threshold	Low	Low	Medium	Medium	High	High
Controls	No	Yes	No	Yes	No	Yes
Observations	36,892	36,892	32,579	32,579	31,542	31,542

Notes. Standard errors (in parentheses) are robust and clustered by municipality. Controls: Altitude, Location, Surface Area, Industry, Assets, Financial leverage, and Return on Investment. All coefficients are multiplied by 10.

** $p < 0.05$; *** $p < 0.01$.

Figure 4. Trends in Inventory-to-Assets, Inventory, and Assets



Note. We plot the trends using three war thresholds: five deaths per 100,000 inhabitants (i.e., a low threshold); 25 deaths per 100,000 inhabitants (i.e., a medium threshold); and 50 deaths per 100,000 inhabitants (i.e., a high threshold).

much longer than the transportation lead times faced by downstream firms. For instance, although a textile retailer can easily adjust the number of garments it will order the next week, a cotton farmer can only adjust inventory by changing the amount of cotton grown. And, because the average cotton boll takes 25 weeks to ripen, inventory decisions only materialize every six months. Given this constraint, a commodity producer can't do anything except plan the

next production batch (whereas downstream firms can adapt swiftly to war-related risk).

The difference in reaction magnitude, between upstream and downstream enterprises, is likely driven by two factors. First, commodity producers have production cycles characterized by high fixed costs because they have to commit labor, land, and capital for many months. But they also enjoy large-scale economies. Thus, although upstream firms find it expensive

Table 8. Model Estimates: Cash Holdings

Specification	I	II	III	IV	V	VI
<i>FARC·AFTER</i>	−0.74** (0.37)	−0.76* (0.42)	−0.95** (0.47)	−0.96** (0.48)	−1.11** (0.50)	−1.12** (0.50)
<i>FARC</i>	2.26*** (0.67)	1.13** (0.45)	2.43*** (0.71)	1.32*** (0.50)	2.53*** (0.75)	1.50*** (0.50)
<i>AFTER</i>	0.85*** (0.21)	0.87*** (0.20)	0.80*** (0.25)	0.82*** (0.23)	0.80*** (0.24)	0.82*** (0.24)
Dependent variable	$\log(\text{cash-to-assets})$					
War threshold	Low	Low	Medium	Medium	High	High
Controls	No	Yes	No	Yes	No	Yes
Observations	36,798	36,798	32,492	32,492	31,455	31,455

Notes. Standard errors (in parentheses) are robust and clustered by municipality. Controls: *Altitude*, *Location*, *Surface Area*, *Industry*, *Assets*, and *Return on Investment*. All coefficients are multiplied by 10.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 9. Model Estimates: Finished, Work-in-Process, and Raw Inventory

Specification	I	II	III	IV	V	VI	VII	VIII	IX
<i>FARC·AFTER</i>	0.10* (0.05)	0.09** (0.04)	0.09** (0.04)	0.25*** (0.08)	0.22*** (0.08)	0.21** (0.09)	0.28*** (0.10)	0.22** (0.09)	0.20** (0.10)
<i>FARC</i>	−0.21*** (0.07)	−0.18** (0.06)	−0.17** (0.07)	0.74*** (0.28)	0.67** (0.27)	0.66** (0.27)	0.76** (0.38)	0.65* (0.35)	0.60* (0.34)
<i>AFTER</i>	0.22*** (0.04)	0.24*** (0.03)	0.24*** (0.03)	−0.30*** (0.04)	−0.30*** (0.04)	−0.30*** (0.04)	−0.31*** (0.06)	−0.27*** (0.06)	−0.27*** (0.06)
Dependent variable	log(<i>finished-inventory-to-assets</i>)			log(<i>work-in-process-inventory-to-assets</i>)			log(<i>raw-inventory-to-assets</i>)		
War threshold	Low	Medium	High	Low	Medium	High	Low	Medium	High
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	23,267	20,381	19,640	36,892	32,579	31,542	36,892	32,579	31,542

Notes. Standard errors (in parentheses) are robust and clustered by municipality. Controls: *Altitude, Location, Surface Area, Industry, Assets, Financial leverage, and Return on Investment*. All coefficients are multiplied by 10.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

to run a new production batch, once they do, they can exploit economies of scale by committing generous land allotments. This means that upstream inventory adjustments span acres worth of product, whereas downstream firms can freely adjust inventories with precision, by the dozen or even by the unit. And second, as Fearon (2005) explains, raw commodities are more vulnerable to conflict because, by their very nature, these inventories are stored in the wild (on farms, vineyards, mines, rigs, and fields), whereas downstream inventories are kept in guarded facilities (alarm-protected counters, vaults, or storage centers).

6.2. Moderator 2: The Market-Access Effect

Firms use distribution hubs—regional markets, distribution centers, seaports, and airports—to acquire inputs and sell output. For this reason, Amodio and Di Maio (2017) find that war wrecks production by restricting access to these hubs. But the closer the firm is located to a distribution hub, the less distance inventory needs to travel, and, all else being equal, the less it will be exposed to road brigands.

To test how this factor moderates the relationship between war and inventory, we first locate all of

Colombia's distribution hubs, using the University Los Andes Geographic Database. We then measure the distance between a firm and the closest hub and produce the following triple diff-in-diffs model:

$$\begin{aligned}
 INV_{it} = & \alpha_t + \beta FARC_i + \psi AFTER_t + \gamma DIST_i \\
 & + \tau(FARC_i \cdot AFTER_t) + \eta(DIST_i \cdot FARC_i) \\
 & + \phi(DIST_i \cdot AFTER_t) + \varphi(AFTER_t \\
 & \cdot FARC_i \cdot DIST_i) + \omega X_{it} + \varepsilon_{it},
 \end{aligned} \quad (7)$$

where $DIST_i$ is the logarithm of the distance between firm i 's location and the nearest hub.⁹

Table 12 shows that the diff-in-diffs coefficient is negative and statistically significant. This means that firms that locate near a distribution hub will reduce inventory during war, whereas far-flung firms will barely do so. In particular, an extra 100 km of distance from a hub lessens the effect by 4.8%–5.3%.¹⁰

This finding can be explained by an economic order quantity (EOQ) model, which benchmarks inventory decisions as a ratio of two factors: (i) the carrying cost (which is inversely proportional to average inventory) and (ii) the acquisition cost (which is directly proportional to average inventory). If a firm is 1 km

Table 10. Supply Chain Analysis

Specification	Raw materials	Manufacturing	Wholesale	Retail
<i>FARC·AFTER</i>	1.32** (0.48)	1.06** (0.44)	1.17*** (0.47)	1.09** (0.50)
<i>FARC</i>	1.08 (1.00)	0.88 (0.73)	0.36 (0.64)	0.84 (1.04)
<i>AFTER</i>	0.40 (0.330)	0.37 (0.22)	0.44* (0.24)	1.27*** (0.40)
Dependent variable	log(<i>inventory-to-assets</i>)			
Controls	Yes	Yes	Yes	Yes
Observations	30,880	25,894	13,316	13,463

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 11. Supply Chain Analysis: Year-by-Year Effects

Specification	One year after	Two years after	Three years after
<i>Raw materials</i>	0.23 (0.17)	0.75*** (0.07)	1.04*** (0.04)
<i>Manufacturing</i>	0.67*** (0.17)	0.31*** (0.16)	0.35*** (0.19)
<i>Wholesale</i>	0.66*** (0.21)	0.42*** (0.19)	0.18 (0.21)
<i>Retail</i>	0.80*** (0.29)	0.08 (0.33)	0.24 (0.38)
Dependent variable	log(<i>inventory-to-assets</i>)		
Controls	Yes	Yes	Yes
Observations	54,181	13,301	12,523

Notes. Standard errors (in parentheses) are robust and clustered by municipality. For illustration purposes, we only display the treatment effect coefficient by year. Controls: *Altitude, Location, Surface Area, Industry, Assets, Financial leverage, and Return on Investment*. All coefficients are multiplied by 10.

*** $p < 0.01$.

away from a distribution hub, transporting inventory is straightforward. Thus, war increases only the firm's carrying cost (the EOQ denominator), but not its acquisition cost (the EOQ numerator), intensifying the inventory reduction. But if the firm is 100 km away from a distribution hub, inventory is vulnerable to road attacks. Thus, war affects both the firm's acquisition cost and carrying cost (the EOQ numerator and denominator), thereby lessening the inventory reduction.

This result can also be explained by changes in the risk perception of firms. This is because distribution hubs are located in areas of commercial interest and, thus, are more susceptible to intense violence. In this fashion, distance could proxy for a firm's perception of the threat of being attacked, which, in turn, can moderate the inventory adjustment.¹¹

7. Extension II: Attack-Level Moderators

Firms are heterogeneous. But so is violence. Our data set is a chronicle of attacks on both private and public infrastructure, of bridge bombings, of massacres, and of road blocks. And each type of attack leaves its own footprint on firms. For instance, when rebels assault nearby businesses, firms may perceive more risk in holding inventory, but when they attack roads, they may perceive more risk in transporting inventory. These mechanisms suggest that the nature of the attack moderates how firms react to violence.

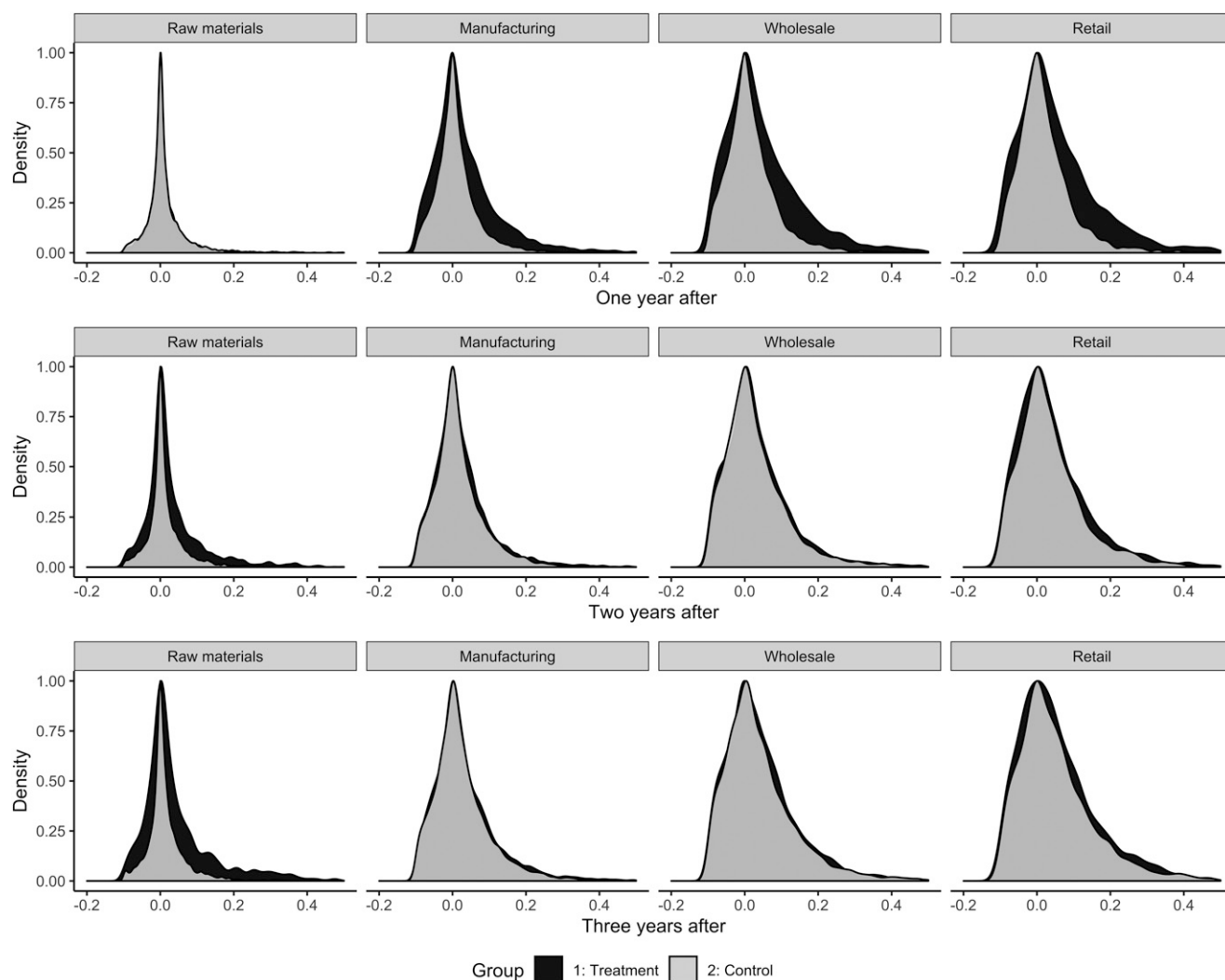
To identify these mechanisms, we develop an identification strategy that examines the moderating effect of different types of attacks. To this end, we follow four steps. First, we identify the date and coordinates of all 5,138 attacks; second, we identify all firms near each attack; third, we measure how these firms adjusted inventory after the attack; and fourth, we benchmark this adjustment against the adjustment made by

almost-identical firms (but located far from any recent attack). As a case in point, FARC attacked cocoa bean regions 23 times during our sample timespan: Magdalena (in Colombia's north) in 2002, Catatumbo (east) in 2007, Valle del Cauca (southwest) in 2009, etc. We identify all adjacent cocoa bean plantations and note how each adjusted inventory after an attack. Then, we compare this change with similar cocoa bean plantations in other regions.

To make these comparisons, we use propensity-score matching, an approach that we describe step by step in Online Appendix III. In summary, this approach couples nearly identical firms by drawing from a treatment and a control group. A firm-year observation is in the treatment group if it witnessed an attack within a 50-km radius in the preceding year; otherwise, it is in the control group. To obtain each firm's propensity score, we use a parametric generalized linear model that estimates approximate standard errors on the treatment effects. This model uses 12 continuous variables, which describe the firm's assets, revenue, income, cost structure, financial leverage, working capital, investment figures, and payroll size, along with binary variables to take into account the firm's industry type. Once we have obtained the propensity scores, we reweigh the distributions and balance them out to calculate the average treatment on the treated (ATT) and average treatment effect (ATE).

Table 13 shows the ATT and ATE estimates. For robustness, we also present estimates from two alternative matching techniques—namely, (i) nonparametric kernel matching and (ii) k -nearest-neighbor matching (where we let $k = 1, 2, 3, 4, 5$, and 6).

All estimates coincide and are significant at the 1% level. In particular, these estimates show that firms significantly decrease inventory when their region is

Figure 5. Supply Chain Effects of War: Inventory Adjustment

Notes. These graphs represent the distribution of inventory adjustment. The x axis represents the annual inventory change ($\Delta_t \text{INV}_t$), in percentage points, for the treatment and control groups. These distributions are decomposed across (i) the supply chain sectors (raw material extraction, manufacturing, wholesale, and retail) and (ii) the posttreatment year (one year, two years, and three years after the treatment date).

subject to attacks. Also, our difference-in-differences and attack-level analyses validate each other. The difference-in-differences estimates show that a *decrease* in conflict intensity causes firms to *increase* inventory, whereas the propensity-score matching estimates show that an *increase* in conflict intensity causes firms to *decrease* inventory.

But how does this inventory reduction depend on the type of attack? To answer this question, we first note that our database divides attacks into four broad categories: attacks on civilians (12%), attacks on private property (18%), attacks on public infrastructure (43%), and attacks on roads and cargo (27%). We disaggregate the estimates across these categories by estimating heterogeneous treatment effects.

Table 14 shows that managers drastically reduce inventory when the rebels attack private property and

civilians, but stockpile when the rebels attack public infrastructure and roads. Again, the EOQ model explains these results. When rebel forces assault nearby factories or private businesses, firms perceive more risk in holding inventory—and, thus, reduce their stock. But when the rebels attack roads and public infrastructure, the firms perceive more risk on shipping inventory—and, thus, increase their stock.

We also measure the financial losses (in Colombian currency) and the human losses (in terms of victims) inflicted by an attack and then analyze how firms react to the magnitude of these losses. Table 14 shows that firms are extremely sensitive to the property losses caused by the attack—the larger the economic damage caused by the attack, the more the firms reduce inventory. These firms, however, are insensitive to the number of casualties. Simply put, operations managers

Table 12. The Market Access Effect: Distance from Distribution Hubs

Specification	I	II
<i>FARC-AFTER-DIST</i>	−0.52** (0.24)	−0.47** (0.24)
<i>FARC-AFTER</i>	0.14 (0.35)	1.07** (0.48)
<i>DIST-FARC</i>	3.49*** (1.23)	3.17*** (1.17)
<i>DIST-AFTER</i>	−0.07 (0.56)	−0.06 (0.72)
<i>FARC</i>	−1.10 (0.72)	−1.11 (0.72)
<i>AFTER</i>	0.02 (0.17)	0.03 (0.48)
<i>DIST</i>	0.44 (2.75)	0.50 (3.45)
Dependent variable	log(<i>inventory-to-assets</i>)	
Controls	No	Yes
Observations	36,892	36,892

Notes. Standard errors (in parentheses) are robust and clustered by municipality. The variable *Dist* is log-linearized. Controls: *Altitude*, *Location*, *Surface Area*, *Industry*, *Assets*, *Financial leverage*, and *Return on Investment*. All coefficients are multiplied by 10.

** $p < 0.05$; *** $p < 0.01$.

think about violence through the lenses of a balance sheet (if the goal of the rebels is to create carnage in its population, then firms are least concerned about being a direct target). And so, whereas they are determined to reduce inventory when they perceive that violence threatens their inventories, they are impervious to war when it just menaces civilians.

8. Robustness Tests

8.1. Spillovers-Robust Specification

Our diff-in-diffs estimates rely on the Stable Unit Treatment Value Assumption (SUTVA), which, in our context, means that the peace process should not meaningfully affect firms in ELN-dominated territories. But the SUTVA is tenuous in our model due to two potential sources of spillovers:

Geographic spillovers. If an ELN-occupied town is 5 km from FARC territory, it would be disingenuous to assume that the effect of the peace process vanishes

at the border without affecting neighboring towns. Thus, the SUTVA is likely contravened when a firm in the control group is adjacent to a FARC-dominated region. In Figure 6 (left), for example, firms B and C are highly susceptible to geographic spillovers, whereas A and D are less vulnerable to these spillovers (because they are far from FARC regions).

Demand spillovers. All other things being equal, firms prefer to source from suppliers located in peaceful territories, which tend to be more stable. Thus, in the aftermath of the peace process, firms in FARC territory enjoyed a comparative advantage, allowing them to wrest demand from firms in ELN territory. Demand spillovers affect firms that produce goods found in both FARC and ELN regions, but are unlikely to affect firms producing goods that are found in only one of the two regions. Consider Figure 6 (right): Firms A, B, and C are vulnerable to demand spillovers because they produce bananas, coffee, and cotton (commodities found in both territories), whereas firm D isn't because it produces sugar cane (a commodity found only in ELN territory).

To deal with this issue, Clarke (2017) develops an identification method that retrieves unbiased diff-in-diffs estimates when such violation happens. We obtain diff-in-diffs estimates that are robust to SUTVA violations via Clarke's method by specifying the following regression equation:

$$\begin{aligned}
 INV_{it} = & \alpha_t + \underbrace{\beta FARC_i}_{\text{treatment indicator}} + \underbrace{\rho AFTER_t}_{\text{posttreatment indicator}} \\
 & + \underbrace{\tau(FARC_i \cdot AFTER_t)}_{\text{diff-in-diffs indicator}} + \underbrace{\omega X_{it}}_{\text{controls}} + \underbrace{\phi R(i, t)}_{\text{spillover indicators}} \\
 & + \underbrace{\mathbb{E}[\eta_i | Farc_i \cdot After_t, R(i, t)]}_{\text{spillover unobserved effect}} + \varepsilon_{it},
 \end{aligned} \tag{8}$$

Table 13. Propensity Score Matching Results

Specification	I	II	III	IV	V	VI	VII	VIII
<i>ATE</i>	−0.76*** (0.22)	−1.95*** (0.60)	−1.09*** (0.61)	−1.10*** (0.61)	−1.50*** (0.61)	−1.51*** (0.61)	−1.52*** (0.61)	−1.64*** (0.61)
<i>ATT</i>	−0.72*** (0.22)	−2.06*** (0.62)	−1.58*** (0.76)	−1.69*** (0.70)	−1.69*** (0.67)	−1.78*** (0.66)	−1.60*** (0.65)	−1.70*** (0.65)
Dependent variable	log(<i>inventory</i>)							
Matching method	PSM	Kernel	NN(1)	NN(2)	NN(3)	NN(4)	NN(5)	NN(6)
Untreated observations	90,545	90,545	90,545	90,545	90,545	90,545	90,545	90,545
Treated observations	41,716	41,716	41,716	41,716	41,716	41,716	41,716	41,716

Notes. NN(k), nearest-neighbor matching using k neighbors; PSM, propensity score matching. In nearest-neighbor matching, we used a caliper of 0.1, meaning that untreated and treated observations are only matched when the distance between the matches is under the caliper threshold. Standard errors (in parentheses) are robust and clustered by municipality. All coefficients are multiplied by 10.

*** $p < 0.01$.

Table 14. Results by Attack Category and Attack Magnitude

Category	I	II	Magnitude	I	II
Civilians	−1.89*** (0.21)	−3.78*** (0.06)	Property Value	−0.50*** (0.15)	−0.54*** (0.15)
Private property	−1.16*** (0.17)	−1.14*** (0.02)	Victims	0.54 (0.03)	0.29 (0.35)
Public infrastructure	0.83*** (0.15)	0.71*** (0.16)			
Roads and cargo	0.09*** (0.01)	0.05*** (0.01)			
Dependent variable	log(inventory)		Dependent variable	log(inventory)	
Method	PMS	NN(1)	Method	PMS	NN
Observations	102,877	102,877	Observations	42,502	42,380

Notes. Standard errors (in parentheses) are robust and clustered by municipality. All coefficients are multiplied by 10. NN(k), nearest-neighbor matching using k neighbors; PSM, propensity score matching.

*** $p < 0.01$.

where $R(i, t)$ is a metric that accounts for spillover effects. We let

$$R(i, t) = \underbrace{\sum_{b=0,25,50} \psi_{geo,d} dist_{geo(b,(b+25))}(i, t)}_{\text{distance spillovers}} + \underbrace{\psi_{indus} dist_{indus}(i, t)}_{\text{demand spillovers}}$$

The dummy variable set $\{dist_{geo(0, 25]}, dist_{geo(25, 50]}, dist_{geo(50, 75]}\}$ measures the geographic distance from firm i to a FARC-dominated territory. For instance, $dist_{geo(25, 50]} = 1$ if observation i is between 25 and 50 km from the closest FARC jurisdiction (and zero otherwise). And dummy $dist_{indus}$ equals one if a firm in an ELN-dominated region produces the same commodity as a firm in a FARC-dominated region (e.g., coffee plantations) and zero if the company does not have competitors in FARC-dominated territory (e.g., sugar-cane plantations).

We present our diff-in-diffs estimates in Table 15, columns (I), (II), and (III), for three distinct specifications. We find that the spillover effects are statistically significant, but also that our diff-in-diffs coefficients are practically unaffected.

8.2. Firms in Peaceful Territories

We now incorporate into our analysis an alternative control group: firms in peaceful territories. First, we run a placebo regression, pretending that firms in peaceful territories were “treated,” as opposed to those in FARC-dominated territories. We tag firms in peaceful territories with the dummy $PEACE$ and use the following specification:

$$INV_{it} = \alpha_t + \rho AFTER_t + \beta PEACE_i + \tau(PEACE_i \cdot AFTER_t) + \delta X_{it} + \varepsilon_{it}. \quad (9)$$

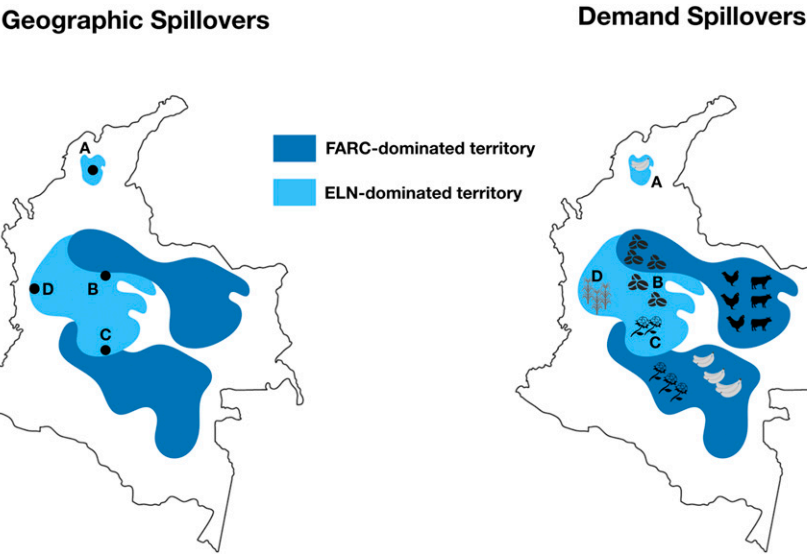
The estimates are presented in Table 16, columns (I)–(III). Next, we benchmark the trends of three groups against each other: (i) firms in FARC-dominated regions, (ii) firms in ELN-dominated regions, and (iii) firms in peaceful regions. For consistency, we let the ELN group be the excluded category in the regression equation:

$$INV_{it} = \alpha_t + \rho AFTER_t + \beta_{FARC} FARC_i + \beta_{PEACE} PEACE_i + \tau_{FARC}(FARC_i \cdot AFTER_t) + \tau_{peace} \times (PEACE_i \cdot AFTER_t) + \delta X_{it} + \varepsilon_{it}. \quad (10)$$

The corresponding estimates are shown in Table 16, columns (IV)–(VI). From these two regressions, we can see that firms in peaceful territories—just like those in ELN regions—are not affected by the treatment, whereas the FARC coefficient (τ_{FARC}) remains statistically significant after including this subsample.

Despite these results, we should be careful when drawing conclusions based on the magnitude of the estimates obtained in this robustness analysis. Whereas firms in ELN and FARC regions are similar, on average, this isn’t the case with firms in peace regions. Peaceful regions comprise mainly urban clusters—like Bogota, Medellin, or Cartagena—with economies that rely on IT-intensive sectors, multinational corporations, tourism, and white-collar jobs. In these clusters, the economy is sophisticated and tech-driven, the population is relatively affluent, and the municipalities are densely packed. In contrast, both FARC- and ELN-dominated war regions comprise small towns and rural economies with a different industry composition. In fact, a test of sample differences, in Tables A.20 and A.21 in the online appendix, shows that FARC and ELN regions are statistically indistinguishable in terms of per capita economic output, population, and industry composition. This is not the case, however, across firms in peaceful territories.

Figure 6. (Color online) Illustration of Supply Chain Spillovers



8.3. The Effect of the Peace Process on Demand

It is conceivable that in the postpeace agreement era, the demand faced by firms in FARC regions incurred a positive uplift (either through exports or growth in the economy of the local region). We obtained data on firms’ incoming orders and examined how these orders changed prepeace agreement and postpeace agreement. Table 17 shows that in the three years after the shock, the peace process had a slightly positive (but statistically insignificant) effect on the incoming orders of all firms.¹² Thus, changes in demand do not explain the inventory shift. Moreover, we analyzed the impact of demand across all supply chain sectors without finding a significant difference across any

sector. The long-term impact of the peace process on demand, however, is likely to be more severe. An interesting avenue of research would be to examine how demand and inventory interact in the long term in reaction to violence swings.

8.4. Other Robustness Tests

We present six additional tests in Online Appendix I. In these tests, we (i) control for the effect of firm entry, relocation, and firm exit; (ii) extend the pretreatment period; (iii) dissect the sample by conflict intensity; (iv) run alternate specifications using unwinsorized samples, fixed and random effects, and different clustering units (from the municipality level to industry and firm level); (v) employ different dependent variables; and (vi) run a cross-sectional placebo test. These tests show that our insights hold across these specifications, meaning that our results are not an artifact of the modeling constructs chosen in the base model.

Table 15. Diff-in-Diffs with Spillovers

Specification	I	II	III
FARC-AFTER	1.09** (0.47)	1.15*** (0.34)	1.11** (0.47)
FARC	−5.61*** (1.10)	1.15 (1.64)	−5.56*** (1.11)
AFTER	−1.29*** (0.28)	−1.33*** (0.26)	−1.29*** (0.29)
dist _{geo} (0,25]	−0.80*** (0.09)		−0.79*** (0.09)
dist _{geo} (25,50]	−0.81*** (0.15)		−0.80*** (0.14)
dist _{geo} (50,75]	−1.53*** (0.08)		−1.49*** (0.08)
dist _{indus}		0.39*** (0.15)	0.35** (0.15)
Dependent variable	log(inventory-to-assets)		
Controls	Yes	Yes	Yes
Observations	36,892	36,892	36,892

p < 0.05; *p < 0.01.

9. Conclusions

9.1. Why Is Civil War Important for Operations Managers?

Researchers study how geopolitical threats affect operations, focusing on natural disasters (Carvalho et al. 2016), climate change (Cachon et al. 2012), terrorism (Cui et al. 2018), and currency fluctuation (Kazaz et al. 2005). But with its emphasis on developed economies, this research stream has overlooked an important source of logistical turmoil: civil wars. As O'Marah (2017) puts it:

Supply chain risk management has enjoyed a rise in prominence over the past decade as events like Japan’s tsunami, Thailand’s floods, and the US west coast port strike. ... Oddly, however, concerns about war remain largely off the radar. Are we fooling ourselves?

Table 16. Firms in Peaceful Territories

Specification	I	II	III	IV	V	VI
<i>FARC-AFTER</i>				0.79** (0.34)	0.91*** (0.31)	1.02*** (0.35)
<i>FARC</i>				−0.28 (0.87)	0.01 (0.91)	0.37 (1.09)
<i>AFTER</i>	−1.37*** (0.23)	−1.37*** (0.23)	−1.29*** (0.29)	−1.38*** (0.21)	−1.40*** (0.20)	−1.33*** (0.24)
<i>PEACE-AFTER</i>	0.21 (0.24)	0.21 (0.24)	0.12 (0.30)	0.62 (0.58)	1.43 (0.54)	−0.39 (0.38)
<i>PEACE</i>	−0.55 (3.06)	0.88 (2.92)	0.35 (3.18)	−4.15** (1.71)	−1.55** (0.61)	−1.23 (0.90)
Dependent variable	log(<i>inventory-to-assets</i>)					
War threshold	Low	Medium	High	Low	Medium	High
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	110,205	110,131	107,810	57,123	71,151	57,123

Notes. Standard errors (in parentheses) are robust and clustered by municipality. Controls: *Altitude*, *Location*, *Surface Area*, *Industry*, *Assets*, *Financial leverage*, and *Return on Investment*. All coefficients are multiplied by 10.

** $p < 0.05$; *** $p < 0.01$.

A constellation of factors makes war a prime geopolitical risk. First, wars are *frequent*. In 2017, civil warfare embroiled a third of the world's population, and, since 1945, there have been 259 conflicts and an annual average of 3,854 war-related attacks according to the Global Terrorism Database. Second, wars are *long-lasting*. The average civil conflict lasts 1,665 days, meaning that, unlike a natural disaster or a terrorist attack, conflicts will disrupt supply chains for years. And, third, wars are *costly*. In 2015, the cost of war amounted to 13% of the world's production by virtue of bombings, assaults on commercial property, road-blocks, loss of physical and human capital, and strikes on public infrastructure.

But war should also be on the radar of operations managers because civil conflicts give rise to a *unique* type of risk. For one thing, there's the fact that warfare puts inventories in the spotlight. Although tornadoes and hurricanes don't actively scout for inventories, insurgents need provisions to subsist, and, thus, prey and thrive on business inventories. Accordingly, they create (rather elaborate) strategies for seizing these inventories. If a firm attempts to store them, marauders will find a way to take them from its factory, and if a firm attempts to transport them, brigands will find a way to snatch them on the road. Rebels also sabotage inventories by hijacking business operations, triggering chaos, intimidating workers, and crippling roads.

For this reason, firms react to conflict-related risk unlike they do to other risks. During interviews, we learned about the risk-management strategies used by managers in war-torn regions. Armed violence, for instance, prompts these managers to avoid using ground routes and, instead, to transport their inventories by air or via safer (but inefficient) routes. It

also prompts them to change the frequency at which inventory travels and to hire security personnel to guard these inventories in camouflaged warehouses. Moreover, war limits the degree of oversight from high-level management and skilled workers (highly paid workers are vulnerable to kidnapping and often have to manage production remotely), which makes it difficult for firms to monitor their yield properly. Thus, taken together, these examples reflect that managers use a unique and creative combination of strategies to respond to the particular set of risks that war brings.

9.2. Managerial and Policy Implications

To effectively deal with war-related turmoil, managers should first be aware of the (rather unusual) mechanisms through which war wreaks havoc on the supply chain. As we show, civil war spurs businesses to replace inventory with cash—oversecuring their financial buffer on the one hand, straining their operational buffer on the other. In the process of reducing inventories, these firms mainly cut their unprocessed inventories, not their finished inventories. What does this mean? It means that, although these firms remain capable of fulfilling planned orders, they become unresponsive to unplanned, or last-minute, orders—a situation that makes their supply chain unfit to deal with demand variability.

Grasping this principle is important in itself, as it can allow policymakers to draft the right policies and funnel the right resources to help war-vulnerable supply chains. For example, injecting liquidity into these supply chains is a poor strategy because war does not create cash constraints; in fact, it does the opposite. Policymakers should, instead, focus on protecting the Achilles heel of war-torn supply

Table 17. Demand Analysis

Specification	All	Commodities	Manufacturing	Wholesale	Retail
<i>FARC·AFTER</i>	0.89 (1.84)	−1.85 (25.71)	0.11 (0.89)	−044 (0.93)	2.04 (10.52)
<i>FARC</i>	−3.37* (1.94)	6.54 (41.39)	−27.21* (14.38)	−21.01 (24.93)	−36.94*** (10.26)
<i>AFTER</i>	−0.64 (1.731)	−22.50 (21.09)	51.91*** (6.58)	56.73*** (8.45)	62.93*** (6.72)
Dependent variable	log(<i>orders</i>)				
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes
<i>Observations</i>	83,553	30,880	25,894	13,316	13,463

Notes. Standard errors (in parentheses) are robust and clustered by municipality. Controls: *Altitude*, *Location*, *Surface Area*, *Assets*, *Financial leverage*, and *Return on Investment*. All coefficients are multiplied by 10.

* $p < 0.10$; *** $p < 0.01$.

chains—namely, on minimizing the impact of unexpected demand swings.

But exactly how can managers protect their supply chains from war-related risk? First, they should identify which inputs are sourced from regions currently experiencing civil war (or likely to be stricken by war).¹³ And, once identified, these managers should adopt the following policies on such inputs:

1. *Redoubling forecasting efforts.* Demand forecasting is always important, but it is exceedingly important when the goods are sourced from a war-ravaged region. Why? When sourcing from stable regions, forecasting errors are less woeful because suppliers usually stockpile and hold spare unprocessed inventories in anticipation of these errors. But war-ridden suppliers don't have the luxury of keeping spare inventories to accommodate poor forecasts. Hence, an error in forecasting an input from a war-ridden region is more likely to result in a stockout and disrupt the supply chain. Managers should, accordingly, channel additional forecasting resources toward these goods.

2. *Move away from pull inventory systems (and into coordinated replenishment systems).* Fast shipping networks have popularized pull inventory systems in international supply chains, which allow downstream firms to rely on lean manufacturing concepts. Unfortunately, these systems are tenuous if goods are sourced from a war-vulnerable supplier, because they expose suppliers to on-the-call demand for new inventory. Instead of using these systems, supply chain managers should favor coordinated replenishment systems. Cachon (2004), Dong and Zhu (2007), and Gürbüz et al. (2007) propose viable alternatives, such as long-term horizon push systems or coordinated replenishment policies. These systems foster advanced planning and a disciplined forecasting culture.

3. *Multi-echelon production smoothing.* Sheffi (2005) argues that if an echelon of the supply chain is operationally inflexible in its replenishment capability,

the rest of the chain should offset this weak spot by implementing multi-echelon production smoothing. Although this tactic requires communication platforms along the supply chain, it can be highly effective in reducing undue demand fluctuation (which pressures the war-ridden supplier and, ultimately, disrupts the supply chain).

4. *Large-shipment consolidation.* Governments can stabilize a war-vulnerable supply chain by facilitating shipment-consolidation platforms, escorted by large-scale transportation security. Dror and Hartman (2007) and Büyükkaramikli et al. (2014) propose two methods to integrate a shipment consolidation strategy.

These policies should be *properly timed*. We show that downstream firms react almost immediately to violence, but long-term, it is the upstream businesses that are hit hardest. Thus, stakeholders should design fast-acting policies toward downstream firms, but powerful and forward-thinking policies to protect upstream enterprises.

These policies should also be *properly tailored*. In general, stakeholders should develop an SAP policy—that is, a policy that incorporates three factors: supply chain position, attack type, and proximity to distribution hubs. For example, to prevent a large inventory reduction from commodity producers, policymakers should tackle the drivers of this behavior—namely, high fixed production costs. On the other hand, firms will stockpile during violent times if they operate far from distribution hubs or if their region is battered with road attacks or public infrastructure strikes. Thus, firms in far-flung, war-ridden corners are less likely to stockout, but are more vulnerable to, and will take a harder hit from, on-the-road attacks. In remote areas, accordingly, policymakers should focus on promoting faster inventory rotation by reducing inventory-transportation risk. Understanding these precise differences will lead us to design the right policy for the right firm at the right time.¹⁴

9.3. Generalizability of Our Results

All wars are different—some are kindled by ethnic enmity, some spring up amid religious bickering, and others sprout up out of territorial ambition. In light of this heterogeneity, is it reasonable to extrapolate our results to other wars? Could we, for instance, compare the seemingly interminable Colombian insurgency war with the short-lived Nigerian Biafran separatist conflict?

In Section 2, we set out some boundaries. We explained how wars are classified based on three characteristics and noted that the Colombian war is recognized as *civil*, *limited*, and *asymmetric*. Moreover, during the sample timespan, the intensity of the Colombian conflict was in the medium-high range. This means that our study can be extrapolated to understand conflicts with similar typology and intensity. Table A.18 and Figure A.6 in the online appendix classify conflicts across these dimensions, indicating that the Colombian civil war is similar to the Libyan Civil War, the Punjab Insurgency War, and the Second Chechen War (but different from the Heglig Crisis War).

Despite these boundaries, the nature of the Colombian civil war allows us to obtain highly generalizable results. For one thing, the Colombian territory is heterogeneous: deserts in the north, mountains in the center, and forests in the south. And the intensity of the conflict is also diverse. In Colombia's south, the number of battle-related deaths was as high as in Syria; in the north, the intensity of the conflict was similar to the one in Nigeria; and in the Pacific coast, the death rate resembled Yemen's. Moreover, rebel groups use similar tactics across the world. The Global Terrorism Database, which catalogs 180,000 attacks across hundreds of wars, shows that rebel groups worldwide act in ways that largely resemble the Colombian groups. The Colombian conflict has generated hundreds of vehicle-burning and cargo-stealing attacks, much like those observed in India, Iraq, and El Salvador. It has also given rise to hundreds of assaults on public infrastructure, much like those in Pakistan, the Philippines, and Thailand. And it has resulted in tens of thousands of attacks on civilians and factories, much like those in Cameroon, Myanmar, and Somalia. Thus, the strategies adopted by Syrian rebels are similar to those adopted by Cameroonian rebels, which, in turn, are similar to those adopted by Colombian rebels—because even if war has many faces, it punches with the same fist.

Endnotes

¹ Table A.18 in the online appendix classifies different wars across these categories.

² Collier's analysis focuses on correlational shifts in aggregate economic production (which is a function of multiple factors, including labor, capital, and technological shifts). By using fine-grained data from firms, our study shows that—in addition to a mere reallocation

in economic output—war imposes a significant reduction on firm-level inventory (and a subsequent increase in fungible assets).

³ See, for example, Cachon and Terwiesch (2008).

⁴ Shortage costs are only factored into inventory models with stochastic demand—when demand is deterministic, this component becomes trivial (Zipkin 2006, chapters 7 and 12). In Section 9, we estimate the effect of the peace process on demand.

⁵ Note that the model-free nature of the diff-in-diffs model lets us obtain consistent estimates without these controls. However, as Angrist and Pischke (2008) argue, including this vector might be required to satisfy the parallel-trends assumption (an assumption that we test in Online Appendix II).

⁶ Note, $100\% \cdot (e^{0.92/10} - 1) \approx 9.6\%$ and $100\% \cdot (e^{1.14/10} - 1) \approx 12.1\%$.

⁷ As in the baseline model, we estimate the dependent variables using a scale-normalized dependent variable.

⁸ We standardize all estimates, in mean and standard deviation, to make the magnitudes comparable.

⁹ Note that our results include dummies to account for the firm's industry, meaning that our results aren't confounded by this factor.

¹⁰ Note, $(e^{0.47/10} - 1) \cdot 100\% \approx 4.8\%$ and $(e^{0.52/10} - 1) \cdot 100\% \approx 5.33\%$.

¹¹ We thank an anonymous reviewer for suggesting this explanation.

¹² The parallel-trends assumption tests for this analysis can be found in the online appendix.

¹³ The Global Conflict Risk Index provides a one- to four-year statistical risk assessment of violent conflict across regions (see <https://resourcewatch.org/>). Managers could lever this index to calculate the potential for each of their suppliers to be affected by war.

¹⁴ With that said, these three factors should be interpreted as comparative statistics, and, without further knowledge on how they interact, one should be careful in combining these effects when drafting a policy.

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